DATA ITEM DESCRIPTION

Title: **THEORY OF EQUIPMENT OPERATION (TEO)**

Number: DI-ADMN-81598
Approval Date: 5 February 2001
AMSC Number: G7414
Limitation: N/A
DTIC Applicable: N/A
GIDEP Applicable: No

Office of Primary Responsibility: G/C12

Applicable Forms: None

Use/relationship:
The Theory of Equipment Operation (TEO) is a report providing information about the architectural design of a system. It describes the top level requirements of the system, the functions of the system, the physical configuration of the system, and it identifies the features of the system satisfying each security requirement. The TEO shall answer four basic questions about the system: 1) What are the top level requirements? 2) What functions are to be designed into the system? 3) Where are the functions physically performed? and 4) What specific design features will satisfy each security requirement?

Requirements:
1. Format. The Theory of Equipment Operation should be in contractors format with the following exceptions:

1.1 Page size. The size of each finished page shall be on 8 1/2 x 11 paper (metric size A4). Drawing and illustrations fold-outs may be used, but must not exceed the 8 1/2 x 11 limits when folded. Photo reduction of oversized pages is preferred, provided such reductions are easily readable and reproducible.

1.2 Binding. The Theory of Equipment Operation should be bound in such a manner that pages can be removed without damage or mutilation.

1.3 Change/revision identification. Changes shall be by replacement change pages with revisions changes and additions from earlier versions being identified by a solid vertical bar and applicable revision number/letter on the outside page margin. Revision deletions shall be identified by an asterisk and revision code in the outside margin.

1.4 Paragraph identification. Each paragraph shall have a unique contractor specified paragraph identifier.

2. Content. The report shall contain front matter as follows:

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2.1 Cover and title page. The following information shall be included on the cover and title page:

a. Report Title.
b. Date of Issue.
c. Report number/revision number or letter.
d. Contract number.
e. Contractor name and address.
f. Program title, including program name.
g. Security classification, if classified.
h. Distribution statement.

2.2 Revision control page. The revision control page list the following information:

a. Each revision number or letter.
b. Date of each revision.
c. Pages affected by each revision.

2.3 Table of contents. The table of contents shall identify the following:

a. The title of the starting page of each major section and paragraph of the report.
b. The page, identifying number, and title of each drawing, illustration, figure and table.

2.4 Chapters. The report shall contain the following chapters:

2.4.1 General information.

The Theory of Equipment Operation (TEO) shall be divided into four chapters. The first chapter describes the top level requirements of the system, the systems operational environment, and the top level security requirements of the system. The second chapter breaks the system down into functional blocks. The third chapter describes the physical configuration of the system and where each function described in chapter 2 will be performed. And the fourth chapter identifies the design features of the system which will satisfy each security requirement.

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The TEO is intended to answer the what and where questions associated with a preliminary system design. The report shall describe the design architecture in enough detail to allow a reader to be able to make decisions about the adequacy of the design approach. The more detailed how questions describing the specific details of design implementation will be addressed in the associated Theory of Compliance.

The report shall be written so that a reader will be able to trace top level functions to top level requirements, understand the hierarchy of functions that will be performed by the system, be able to identify where each function will be performed in the systems physical architecture, and understand the design approach for satisfying each security requirement identified in the Security Requirements.
Liberal use of diagrams and charts to support the written description is encouraged throughout the TEO.

2.4.2 Chapter 1. System Requirements and Operational Environment:

The first chapter of the TEO is to provide a concise description of the top level requirements or goals of the system design, a description of the operational environment and constraints (e.g. airborne, command post, access by cleared personnel only), and a statement of the top level security requirements or goals. This chapter will tell the reader what the system is supposed to do, in what environment the system must operate, and what the operational security concerns are.

2.4.3 Chapter 2. Functional Architecture:

The second chapter of the TEO is to provide a functional description of the system being designed. It provides the reader with an understanding of the hierarchy of functions within the system and how this functional architecture will satisfy the top level requirements identified in Chapter 1. The description shall logically flow from top level functions, down through several layers of functional partitioning, to a functional design level where each function represents an individual task and can be identified as occurring within or by a specific physical component of the system.

2.4.4 Chapter 3. Physical Architecture:

The third chapter of the TEO is to provide a physical description of the system being designed. It provides the reader with an understanding of the physical components of the system (i.e. hardware, software, databases, communication paths, etc.) their relationships to each other (e.g. Master/Slave, subcomponent, etc.), and it associates the functions identified in Chapter 2 with specific components.

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2.4.5 Chapter 4. Security Architecture:

The fourth chapter of the TEO is to assure that each security requirement/goal identified in the Security Requirements has been individually addressed, and that an adequate design approach to satisfying the requirement has been proposed.

Each requirement in the Security Specification shall be addressed separately and shall describe the design approach proposed for satisfying that requirement. This description shall provide system specific detail on how the design will satisfy the requirement and not just a restatement of the requirement. If that requirement applies to more than one area of the design, the design approach for each area shall be addressed separately.

If during the preliminary design phase the designer believes that a stated security requirement is not applicable to the emerging design approach, a justification for why that requirement is not applicable shall be provided in lieu of the design approach. The purpose of this is to provide a mechanism for security requirements to be further tailored to the design. If the government agrees with the justification, the requirement need not be addressed in subsequent documentation. On the other hand, if the government believes that the requirement is still applicable, the designer is then responsible for providing the government with a description of the design approach for satisfying the requirement as described above.

3. End of DI-ADMN-81598

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